Ternary Qutrit Systems on $\mathbb{C}^{27}$: Universal Generation of $\mathfrak{su}(27)$ from Local $\mathfrak{su}(3)$ Gates and a Single CNOT

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Abstract

We analyze the Lie-algebraic structure of a three-qutrit quantum system whose state space is $\mathbb{C}^{27} = \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$. We prove that local $\mathfrak{su}(3)$ generators on each qutrit together with the ternary CNOT gate on any two wires sharing a common wire generate the full Lie algebra $\mathfrak{su}(27)$ (dimension 728). In contrast, a single CNOT generates only $\mathfrak{su}(9) \oplus \mathfrak{su}(3)$ (dimension 88). The 27 computational basis states form an orthonormal basis of $\mathbb{C}^{27}$; the projective measurement in this basis defines the natural readout. The discrete symmetry group generated by CNOT gates and local involutions is isomorphic to $\mathrm{GL}(3,3)$ (order 11,232), acting transitively on the 26 non-zero basis states ($|000\rangle$ is fixed). These results provide the minimal gate set required for universal quantum computing on $\mathbb{C}^{27}$ using only local $\mathfrak{su}(3)$ operations and ternary CNOT couplings. This provides the minimal gate set for universal quantum computing on three qutrits.

1 Introduction

1.1 Motivation

A qutrit (quantum ternary digit) is a three-level quantum system with state space $\mathbb{C}^3$. Its computational basis states $\{|0\rangle, |1\rangle, |2\rangle\}$ carry $\log_2 3 \approx 1.585$ qubits of information. The tensor product of three qutrits produces a 27-dimensional Hilbert space $\mathbb{C}^{27}$. This raises a natural question: can local unitary gates acting on each qutrit individually, together with a single coupling gate (the ternary CNOT), generate the full special unitary group SU(27)? Or does the system remain confined to a proper subgroup equivalent to three independent qutrits?

Qutrit-based quantum computing has attracted substantial attention in recent years. Metaplectic anyon models based on $\mathrm{SU}(2)_4$ naturally support ternary computational bases, with the CNOT gate arising as a braiding operation [9, 12]. Ternary quantum arithmetic circuits, including ripple-carry adders, carry-lookahead adders, and multipliers, have been developed systematically [10, 11, 13]. The qutrit Clifford group (the normalizer of the generalized Pauli group) plays a central role in quantum error correction [3], and magic state distillation protocols for qutrits are known to offer advantages over qubit-based protocols [4].

This paper addresses the Lie-algebraic foundation of ternary quantum computation. Rather than analyzing the discrete Clifford group, we study the *continuous* unitary transformations generated by local $\mathfrak{su}(3)$ Hamiltonians on each qutrit together with the ternary

CNOT gate. The central result is that two overlapping CNOT gates generate the full Lie algebra $\mathfrak{su}(27)$ (dimension 728), implying that any unitary in SU(27) can be approximated arbitrarily well using this gate set. By the Solovay–Kitaev theorem, the gate set is efficient: any target unitary can be approximated to precision ε using $\mathrm{poly}(\log(1/\varepsilon))$ gates from the finite generating set.

1.2 Summary of Results

Our main results are:

1. **Two-wire universality (Theorem 3).** The Lie algebra generated by local $\mathfrak{su}(3)$ generators on two qutrits and the ternary CNOT gate is $\mathfrak{su}(9)$ (dimension 80). The proof uses Dynkin's classification of maximal subalgebras: $\mathfrak{su}(3)\oplus\mathfrak{su}(3)$ is a Type III maximal subalgebra of $\mathfrak{su}(9)$, and the CNOT gate lies outside the corresponding subgroup $\mathrm{SU}(3)\times\mathrm{SU}(3)$.

2. **Three-wire universality (Theorem 4).** With three qutrits, two CNOT gates sharing a common qutrit generate the full $\mathfrak{su}(27)$ (dimension 728). One CNOT generates only $\mathfrak{su}(9)\oplus\mathfrak{su}(3)$ (dimension 88).

3. **Spanning tree generation (Theorem 7).** For $k \geq 2$ qutrits, CNOT gates on edges forming a spanning tree generate $\mathfrak{su}(3^k)$. The minimal number of CNOT gates required is $k-1$.

4. **Discrete symmetry group (Proposition 9).** The discrete group generated by CNOT gates (both directions) on all qutrit pairs together with the local inversion gates T_{inv} on each qutrit is isomorphic to $\mathrm{GL}(3,3)$ (the general linear group over $\mathrm{GF}(3)^3$), of order 11,232, acting transitively on the 26 non-zero basis states ($|000\rangle$ is fixed).

5. **Orthonormal basis of $\mathbb{C}^{27}$.** The 27 computational basis states form the standard orthonormal basis of $\mathbb{C}^{27}$. Projective measurement in this basis defines the natural readout, with the winner-take-all rule selecting the basis state with the largest post-measurement probability.

1.3 Relation to Prior Work

The study of ternary quantum gates in the metaplectic anyon framework was pioneered by Cui and Wang [12], who showed that the controlled-X gate (the ternary CNOT analog) together with single-qutrit Clifford gates generates the full ternary Clifford group. Bocharov et al. [9] extended this to arithmetic circuits and proved that Clifford + $C(X)$ generates $\mathrm{GL}(n,\mathbb{F}_3)\ltimes\mathbb{F}_3^n$ as a maximal subgroup of the symmetric group.

Dynkin's classification of maximal subalgebras of classical Lie algebras [8] provides the key technical tool for our proof. The tensor product embedding $\mathfrak{su}(m)\oplus\mathfrak{su}(n)\to\mathfrak{su}(mn)$ is a Type III (non-regular non-simple) maximal subalgebra for $m,n\geq 3$. This result is the complex analog of the orthogonal case studied in the classical quantum-mimetic setting [14], and it underpins our universality proof.

Our work complements the discrete Clifford group approach by analyzing the *continuous* reachable set: the full sphere S^{53} in $\mathbb{C}^{27}\cong\mathbb{R}^{54}$ under the realification map. While the Clifford group is finite (the normalizer of the generalized Pauli group), the

continuous $\mathfrak{su}(27)$ contains all Hamiltonians, enabling arbitrary unitary synthesis via the Solovay–Kitaev theorem.

2 State Space

Let $\mathcal{V} = \mathbb{C}^3$ with ordered orthonormal basis

$$\mathcal{B} = \{|0\rangle, |1\rangle, |2\rangle\}, \qquad |0\rangle = \begin{pmatrix}1\\0\\0\end{pmatrix}, \ |1\rangle = \begin{pmatrix}0\\1\\0\end{pmatrix}, \ |2\rangle = \begin{pmatrix}0\\0\\1\end{pmatrix}.$$

Three qutrits $\mathcal{A}, \mathcal{B}, \mathcal{C}$ produce the tensor product space

$$\mathcal{H} = \mathcal{V}_\mathcal{A} \otimes \mathcal{V}_\mathcal{B} \otimes \mathcal{V}_\mathcal{C} \cong \mathbb{C}^{27}.$$

A pure quantum state is

$$|\Psi\rangle = \sum_{i,j,k \in \{0,1,2\}} c_{ijk} \, |i\rangle_\mathcal{A} \otimes |j\rangle_\mathcal{B} \otimes |k\rangle_\mathcal{C}, \qquad \sum_{i,j,k} |c_{ijk}|^2 = 1.$$

The 27 basis vectors form an orthonormal basis of $\mathbb{C}^{27}$. Under realification $\mathbb{C}^{27} \cong \mathbb{R}^{54}$, the unit sphere is S^{53}.

3 Lie Algebra Generation

3.1 Local Generators

The Lie algebra $\mathfrak{su}(3)$ has dimension 8. A standard basis consists of the Gell-Mann matrices $\lambda_1, \ldots, \lambda_8$, normalized so that $\mathrm{tr}(\lambda_a \lambda_b) = 2\delta_{ab}$, or equivalently the anti-Hermitian generators $L_a = i\lambda_a$ satisfying $[L_a, L_b] = 2 f_{abc} L_c$ where f_{abc} are the $\mathfrak{su}(3)$ structure constants.

A convenient choice for our purposes uses the Cartan subalgebra generators

$$H_1 = i \begin{pmatrix}0&0&0\\0&1&0\\0&0&-1\end{pmatrix}, \qquad H_2 = i \begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix},$$

together with the ladder operators

$$E_{12} = \begin{pmatrix}0&1&0\\-1&0&0\\0&0&0\end{pmatrix}, \ E_{13} = \begin{pmatrix}0&0&1\\0&0&0\\-1&0&0\end{pmatrix}, \ E_{23} = \begin{pmatrix}0&0&0\\0&0&1\\0&-1&0\end{pmatrix},$$

and their $\mathfrak{su}(3)$ commutators.

These are lifted to $\mathfrak{su}(27)$ via Kronecker products:

$$L_A(X) = X \otimes I \otimes I, \quad L_B(X) = I \otimes X \otimes I, \quad L_C(X) = I \otimes I \otimes X.$$

For two qutrits, we write

$$L_A(X) = X \otimes I, \qquad L_B(X) = I \otimes X,$$

acting on $\mathbb{C}^9 = \mathbb{C}^3 \otimes \mathbb{C}^3$.

The 24 local generators (8 per qutrit) generate $\mathfrak{su}(3) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(3)$, a 24-dimensional proper subalgebra of $\mathfrak{su}(27)$.

3.2 The Ternary CNOT Gate

The ternary CNOT acts on two qutrits as

$$\mathrm{CNOT} : |i, j\rangle \mapsto |i,\, j \oplus i\rangle,$$

where $\oplus$ denotes addition modulo 3. As a 9×9 permutation matrix, it lies in SU(9) (all eigenvalues are third roots of unity; determinant 1). Explicitly, ordering the basis as $|00\rangle, |01\rangle, |02\rangle, |10\rangle, |11\rangle, |12\rangle, |20\rangle, |21\rangle, |22\rangle$, the CNOT matrix is

$$\mathrm{CNOT} = \begin{pmatrix} I & 0 & 0 \\ 0 & S & 0 \\ 0 & 0 & S^2 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

where S is the cyclic shift on $\mathbb{C}^3$ ($S^3 = I$).

Lifted to $\mathbb{C}^{27}$, the CNOT on qutrits A and B with qutrit C as spectator is

$$\mathrm{CNOT}_{AB}^{(27)} = \mathrm{CNOT} \otimes I \in \mathrm{SU}(27).$$

3.3 Generation Theorem: The Correct Formulation

Let G be the subgroup generated by the local unitary groups $\exp(\mathfrak{su}(3)_A)$, $\exp(\mathfrak{su}(3)_B)$, and the CNOT gate. Formally,

$$G = \langle \exp(\mathfrak{su}(3)_A),\, \exp(\mathfrak{su}(3)_B),\, \mathrm{CNOT} \rangle \subset \mathrm{SU}(9).$$

Let $L = \mathrm{Lie}(G)$ be the Lie algebra of G. Since CNOT is a discrete element, its adjoint action $\mathrm{Ad}_{\mathrm{CNOT}}$ is an automorphism of L. Therefore L is the smallest Lie subalgebra of $\mathfrak{su}(9)$ that contains $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$ and is invariant under $\mathrm{Ad}_{\mathrm{CNOT}}$. Equivalently,

$$L = \left\langle \mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B,\, \mathrm{Ad}_{\mathrm{CNOT}}(\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B),\, \mathrm{Ad}^2_{\mathrm{CNOT}}(\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B), \dots \right\rangle_{\mathrm{Lie}},$$

where $\langle \cdot \rangle_{\mathrm{Lie}}$ denotes Lie closure (repeated commutators). Since $\mathrm{CNOT}^3 = I$, the adjoint action has order 3, so only $\mathrm{Ad}_{\mathrm{CNOT}}$ and $\mathrm{Ad}^2_{\mathrm{CNOT}}$ need be considered.

3.4 Maximal Subalgebra Argument

The key to the proof is the following result from Dynkin's classification of maximal subalgebras of classical Lie algebras, now specialized to Type A ($\mathfrak{sl}(n, \mathbb{C})$, whose compact real form is $\mathfrak{su}(n)$).

Lemma 1 (Dynkin classification)**.** *Let $m, n \geq 3$. The tensor product embedding*

$$\mathfrak{su}(m) \oplus \mathfrak{su}(n) \to \mathfrak{su}(mn)$$

via the irreducible representation $\mathbb{C}^m \otimes \mathbb{C}^n$ is a maximal subalgebra of $\mathfrak{su}(mn)$. In Dynkin's classification [8], this is a non-regular non-simple *subalgebra (Type III) of the Type A series. The embedding corresponds to the branching rule $\mathfrak{su}(mn) \downarrow \mathfrak{su}(m) \oplus \mathfrak{su}(n)$, under which the adjoint representation of $\mathfrak{su}(mn)$ decomposes as*

$$(\mathfrak{su}(m) \oplus \mathfrak{su}(n)) \oplus (\mathfrak{su}(m) \otimes \mathfrak{su}(n)),$$

where $\mathfrak{su}(m) \otimes \mathfrak{su}(n)$ is an irreducible summand (of dimension $(m^2 - 1)(n^2 - 1)$) not contained in $\mathfrak{su}(m) \oplus \mathfrak{su}(n)$. In Dynkin's classification of maximal subalgebras (the tensor-product $A_{m-1} \times A_{n-1} \subset A_{mn-1}$), the embedding $\mathfrak{su}(m) \oplus \mathfrak{su}(n) \subset \mathfrak{su}(mn)$ is maximal for integers $m, n \geq 3$ (up to the obvious order and the standard $(2, 2)$ coincidence which does not arise here); the cases used in this paper are $(m, n) = (9, 3), (3, 3)$.

Proof. See Dynkin [8, §15, no. 46, Type III table (p. 114)], where the non-regular non-simple maximal subalgebras of the classical Lie algebras are tabulated. For $\mathfrak{sl}(N,\mathbb{C})$ (the complexified form of $\mathfrak{su}(N)$), the Type III maximal subalgebras include

$$\mathfrak{sl}(s,\mathbb{C}) \oplus \mathfrak{sl}(t,\mathbb{C}) \subset \mathfrak{sl}(st,\mathbb{C}) \qquad (s,t \geq 2,\ s,t \neq 4).$$

Taking the compact real form gives the $\mathfrak{su}(s) \oplus \mathfrak{su}(t) \subset \mathfrak{su}(st)$ embedding. The representation $\mathbb{C}^m \otimes \mathbb{C}^n$ of $\mathfrak{su}(m) \oplus \mathfrak{su}(n)$ is irreducible, and the dimension count

$$\dim(\mathfrak{su}(m) \oplus \mathfrak{su}(n)) = (m^2-1) + (n^2-1)$$

is strictly less than $\dim \mathfrak{su}(mn) = (mn)^2 - 1$. Maximality follows from Dynkin's Table 12 (Type III row). □

Corollary 2. *The subalgebra $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$ is a maximal subalgebra of $\mathfrak{su}(9)$. It arises from the Type III tensor product embedding with $m = n = 3$.*

3.5 Two-Wire Generation

Theorem 3 (Two-wire generation)**.** *For $k = 2$, the Lie algebra generated by $\mathfrak{su}(3)_A$, $\mathfrak{su}(3)_B$, and CNOT(A,B) (in the sense of Section 3.3) is the full $\mathfrak{su}(9)$ (dimension 80).*

Proof. Let L be the Lie algebra generated by $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$ and CNOT, i.e., $L = \mathrm{Lie}(G)$ where $G = \langle \exp(\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B), \mathrm{CNOT} \rangle$.

First, we show that CNOT $\notin \mathrm{SU}(3) \times \mathrm{SU}(3)$. The subgroup $\mathrm{SU}(3) \times \mathrm{SU}(3)$ consists of matrices of the form $U_A \otimes U_B$ with $U_A, U_B \in \mathrm{SU}(3)$. If CNOT $= A \otimes B$, then in the 3×3 block decomposition (block index from qutrit A), the (i,j)-th block would be $A_{ij}B$. Since CNOT has the block-diagonal form $\mathrm{diag}(I, S, S^2)$ with S the cyclic shift, all off-diagonal blocks vanish. For a non-zero B, this forces $A_{ij} = 0$ for all $i \neq j$, so A must be diagonal: $A = \mathrm{diag}(a_1, a_2, a_3)$. Then CNOT $= \mathrm{diag}(a_1B, a_2B, a_3B)$. Equating blocks gives $a_1B = I$, $a_2B = S$, $a_3B = S^2$. From $a_1B = I$ we obtain $B = a_1^{-1}I$, which substituted into $a_2B = S$ gives $a_2a_1^{-1}I = S$, implying S is a scalar multiple of the identity – a contradiction. Hence CNOT is not a tensor product, and CNOT $\notin \mathrm{SU}(3) \times \mathrm{SU}(3)$.

We show instead that an adjoint conjugate of a local generator lies outside $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$. Let $J_B^z = I \otimes H_1 \in \mathfrak{su}(3)_B$. From the explicit conjugation above,

$$\mathrm{Ad}_{\mathrm{CNOT}}(J_B^z) = \begin{pmatrix} H_1 & 0 & 0 \\ 0 & SH_1S^{-1} & 0 \\ 0 & 0 & S^2H_1S \end{pmatrix},$$

with $H_1 = i\ \mathrm{diag}(0,1,-1)$, hence $SH_1S^{-1} = i\ \mathrm{diag}(-1,0,1)$ and $S^2H_1S = i\ \mathrm{diag}(1,-1,0)$. Any element of $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$ has, in this 3×3 block decomposition, diagonal blocks $X_{ii}I_{\mathbb{C}^3} + Y$ with a common $Y \in \mathfrak{su}(3)_B$ and diagonal entries X_{ii} of $X \in \mathfrak{su}(3)_A$; therefore any two diagonal blocks differ by a scalar multiple of the identity. Here

$$SH_1S^{-1} - H_1 = i\ \mathrm{diag}(-1,-1,2),$$

which is not a scalar multiple of the identity. Thus $\mathrm{Ad}_{\mathrm{CNOT}}(J_B^z) \notin \mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$. Since L is the Lie algebra of a group containing CNOT, it is closed under $\mathrm{Ad}_{\mathrm{CNOT}}$, so L contains this element and strictly contains $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$.

By Corollary 2, $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$ is a maximal subalgebra of $\mathfrak{su}(9)$. Since L strictly contains a maximal subalgebra, we must have $L = \mathfrak{su}(9)$. □

3.6 Explicit Commutator Verification

The maximal subalgebra argument above is sufficient for the proof, but we can also verify directly that the CNOT generator couples the two qutrits. Let

$$J_B^z = I \otimes H_1 \in \mathfrak{su}(3)_B, \qquad J_B^x = I \otimes E_{12} \in \mathfrak{su}(3)_B,$$

and consider the adjoint action of CNOT on J_B^z. Writing CNOT in block form as $\mathrm{diag}(I, S, S^2)$ where S is the cyclic shift, we obtain

$$\mathrm{Ad}_{\mathrm{CNOT}}(J_B^z) = \mathrm{CNOT} \cdot (I \otimes H_1) \cdot \mathrm{CNOT}^{-1} = \begin{pmatrix} H_1 & 0 & 0 \\ 0 & SH_1S^{-1} & 0 \\ 0 & 0 & S^2H_1S \end{pmatrix}.$$

A brief computation shows SH_1S^{-1} and S^2H_1S are distinct $\mathfrak{su}(3)$ elements related by the conjugation action of S, which cyclically permutes the basis. The resulting block-diagonal matrix is not a pure tensor $I \otimes (\cdot)$: it has different 3×3 blocks on each diagonal sector, mixing the A and B degrees of freedom.

3.7 Three-Wire Generation

Theorem 4 (Three-wire generation)**.** *For $k = 3$, the Lie algebra generated by $\mathfrak{su}(3)_A$, $\mathfrak{su}(3)_B$, $\mathfrak{su}(3)_C$, and any two CNOT gates sharing a common wire (e.g., CNOT(A, B) and CNOT(A, C)) is the full $\mathfrak{su}(27)$ (dimension 728). A single CNOT generates only $\mathfrak{su}(9) \oplus \mathfrak{su}(3)$ (dimension 88).*

Proof. The single-CNOT case follows from Theorem 3 on qutrits A, B together with the disjoint-support lemma: a generator on qutrit C commutes with all elements acting only on A, B, so the algebra decomposes as $\mathfrak{su}(9)_{AB} \oplus \mathfrak{su}(3)_C$.

For the two-CNOT case, let L be the generated algebra (in the sense of Section 3.3). By Theorem 3 applied to (A, B) and (A, C), we have $\mathfrak{su}(9)_{AB} \subset L$ and $\mathfrak{su}(9)_{AC} \subset L$. Both contain $\mathfrak{su}(3)_A$ (the local algebra on qutrit A).

Now consider the subalgebra $\mathfrak{su}(9)_{AB} \oplus \mathfrak{su}(3)_C \subset \mathfrak{su}(27)$. By Lemma 1 with $m = 9$, $n = 3$, the embedding $\mathfrak{su}(9) \oplus \mathfrak{su}(3) \to \mathfrak{su}(27)$ via $\mathbb{C}^9 \otimes \mathbb{C}^3$ is a Type III maximal subalgebra of $\mathfrak{su}(27)$.

We claim that $\mathfrak{su}(9)_{AC}$ is not contained in $\mathfrak{su}(9)_{AB} \oplus \mathfrak{su}(3)_C$. Every element of $\mathfrak{su}(9)_{AC}$ has, in the 3×3 block decomposition relative to qutrit A, the form $\sum_{i=0}^{2} |i\rangle\langle i|_A \otimes I_B \otimes M_i$ with $M_i \in \mathfrak{su}(3)$; for instance $G = \mathrm{Ad}_{\mathrm{CNOT}_{AC}}(J_C^z)$ has this form with $M_i = S^i H_1 S^{-i}$, which are distinct for $i = 0, 1, 2$. Suppose such a G also lies in the direct sum, so that $G = H_{AB} \otimes I_C + I_{AB} \otimes K_C$ with $H_{AB} \in \mathfrak{su}(9)$ and $K_C \in \mathfrak{su}(3)$. Taking the partial trace over qutrit C gives $\mathrm{tr}_C(G) = 3H_{AB}$ (the K_C term contributes $I_B \,\mathrm{tr}(K_C) = 0$ because $\mathfrak{su}(3)$ elements are traceless). But since each M_i is traceless, $\mathrm{tr}_C(G) = \sum_i |i\rangle\langle i|_A \otimes I_B \,\mathrm{tr}(M_i) = 0$, so $H_{AB} = 0$. Then $G = I_{AB} \otimes K_C$ has all A-blocks equal to $I_B \otimes K_C$, contradicting the distinct blocks $I_B \otimes M_i$ of G. Hence $\mathfrak{su}(9)_{AC} \not\subset \mathfrak{su}(9)_{AB} \oplus \mathfrak{su}(3)_C$.

Since L contains $\mathfrak{su}(9)_{AB} \oplus \mathfrak{su}(3)_C$ (because $\mathfrak{su}(9)_{AB} \subset L$ and $\mathfrak{su}(3)_C \subset L$) and also contains $\mathfrak{su}(9)_{AC}$ which is not contained in this subalgebra, L strictly contains the maximal subalgebra. Therefore $L = \mathfrak{su}(27)$.

Explicitly, consider the commutator of elements from $\mathfrak{su}(9)_{AB}$ and $\mathfrak{su}(9)_{AC}$. Let $Y = \mathrm{Ad}_{\mathrm{CNOT}_{AC}}(J_C^z) \in \mathfrak{su}(9)_{AC}$ and $X = [J_A^x, \mathrm{Ad}_{\mathrm{CNOT}_{AB}}(J_B^z)] \in \mathfrak{su}(9)_{AB}$. The bracket $[X, Y]$ has components acting on all three qutrits; its partial trace over C vanishes (since Y is

traceless on C) while the full commutator is non-zero, so it cannot lie in $\mathfrak{su}(9)_{AB} \oplus \mathfrak{su}(3)_C$. More concretely, $[X, Y]$ has non-zero norm $\left\|[X, Y]\right\|_{\mathrm{F}} > 0$ and satisfies $\mathrm{tr}_C([X, Y]) = 0$, whereas any element of the direct sum with $\mathrm{tr}_C(\cdot) = 0$ must be pure $\mathfrak{su}(9)_{AB}$, which cannot produce the three-qutrit correlations present in $[X, Y]$. □

Remark 5. *The SWAP gate does not enlarge the algebra: SWAP(A, B) is contained in the Lie closure of the local generators, acting as a discrete Weyl group element.*

3.8 Disjoint Support Lemma

Lemma 6 (Disjoint support)**.** *Let $V = V_1 \otimes V_2$ be a tensor product of finite-dimensional complex vector spaces. For any $X \in \mathfrak{su}(V_1)$ and $Y \in \mathfrak{su}(V_2)$, the lifted generators satisfy*

$$[X \otimes I_2,\ I_1 \otimes Y] = 0$$

in $\mathfrak{su}(V_1 \otimes V_2)$. More generally, if $A \in \mathfrak{su}(V_1 \otimes \cdots \otimes V_k)$ acts non-trivially only on qutrits $S \subseteq \{1, \ldots, k\}$ and B acts non-trivially only on $T \subseteq \{1, \ldots, k\}$ with $S \cap T = \varnothing$, then $[A, B] = 0$.

Proof. Immediate from the definition of the Kronecker product:

$$(X \otimes I)(I \otimes Y) = X \otimes Y = (I \otimes Y)(X \otimes I).$$

The general case follows by embedding each operator into the full tensor product via $A \otimes I_{[k]\setminus S}$ and noting that disjoint supports give commuting factors. □

3.9 Generation Theorem for k Qutrits

Theorem 7 (Spanning tree generation)**.** *For $k \geq 2$, the Lie algebra generated by $\mathfrak{su}(3)_i$ for each qutrit $i = 0, \ldots, k-1$ together with $CNOT_{i,j}$ for edges forming a spanning tree on k qutrits (exactly $k-1$ edges) is the full $\mathfrak{su}(3^k)$.*

Proof. We prove by induction on k. The base case $k = 2$ is Theorem 3.

Assume the theorem holds for $k-1$ qutrits. Let T be a spanning tree on k qutrits with leaf l adjacent to p. Let $V = \mathbb{C}^{3^{k-1}}$ (the state space of the $k-1$ qutrits excluding l) and $W = \mathbb{C}^3$ (qutrit l).

By the induction hypothesis applied to the tree $T \setminus \{l\}$, the Lie algebra $\mathfrak{su}(V)$ is contained in the generated algebra L. By Theorem 3 applied to qutrits p and l, $\mathfrak{su}(9)_{p,l} \subset L$ (in the lifted sense, with all other qutrits as spectators). The intersection $\mathfrak{su}(V) \cap \mathfrak{su}(9)_{p,l}$ contains $\mathfrak{su}(3)_p$ (the local algebra on qutrit p), so both subalgebras are non-trivially coupled.

Now consider the subalgebra $\mathfrak{su}(V) \oplus \mathfrak{su}(W) \subset \mathfrak{su}(V \otimes W) = \mathfrak{su}(3^k)$. The embedding $\mathfrak{su}(V) \oplus \mathfrak{su}(W) \to \mathfrak{su}(V \otimes W)$ via the tensor product $\mathbb{C}^{3^{k-1}} \otimes \mathbb{C}^3$ is a Type III maximal subalgebra by Lemma 1: here $m = 3^{k-1}$ and $n = 3$. Neither m nor n equals 2 or 4 for any $k \geq 2$ (since 3^{k-1} is a power of 3), so the maximality condition holds for all $k \geq 2$.

The generated algebra L contains $\mathfrak{su}(V) \oplus \mathfrak{su}(W)$ (by induction and Theorem 3). The CNOT gate on (p, l) is not contained in $\mathrm{SU}(V) \times \mathrm{SU}(W)$ (it mixes the qutrits non-trivially, as argued in the proof of Theorem 3), so its adjoint action on $\mathfrak{su}(V) \oplus \mathfrak{su}(W)$ generates elements outside the maximal subalgebra. Hence L strictly contains $\mathfrak{su}(V) \oplus \mathfrak{su}(W)$, and by maximality $L = \mathfrak{su}(V \otimes W) = \mathfrak{su}(3^k)$.

This completes the induction. □

Corollary 8. *For $k \geq 4$, two CNOTs on $(0,1)$ and $(0,2)$ generate a proper subalgebra of dimension $80 + 8(k-3)$, leaving qutrits $3, \ldots, k-1$ uncoupled.*

The bound $k-1$ is also necessary: with fewer than $k-1$ CNOT gates, the graph on k qutrits is disconnected. By Lemma 6 (disjoint support), generators acting on disconnected components commute, so the generated Lie algebra decomposes as a direct sum of algebras acting on the tensor factors of each connected component. A direct sum of proper subalgebras cannot equal the simple algebra $\mathfrak{su}(3^k)$. Hence the minimal number of CNOT gates required for full generation is exactly $k-1$.

4 Orbit Structure

The orbit of a state under a Lie subalgebra $\mathfrak{h} \subseteq \mathfrak{su}(27)$ reveals the accessible state space:

Subalgebra	$\dim \mathfrak{h}$	Generic orbit	$\dim_{\mathbb{R}}$
$\mathfrak{su}(3)^3$	24	$\mathbb{C}P^2 \times \mathbb{C}P^2 \times \mathbb{C}P^2$	12
$\mathfrak{su}(9) \oplus \mathfrak{su}(3)$	88	$\mathbb{C}P^8 \times \mathbb{C}P^2$	20
$\mathfrak{su}(27)$	728	$\mathbb{C}P^{26}$	52

Under local gates only, the state is confined to the Segre variety $\sigma_{1,1,1}(\mathbb{C}P^2 \times \mathbb{C}P^2 \times \mathbb{C}P^2)$ of product states, a 12-dimensional real submanifold of $\mathbb{C}P^{26}$. One CNOT expands the accessible space to $\mathbb{C}P^8 \times \mathbb{C}P^2$ (20 real dimensions). Two overlapping CNOTs unlock the full projective space $\mathbb{C}P^{26}$ (52 real dimensions).

This orbit dimension jump is the precise mathematical signature of genuine three-way entanglement. Non-generic orbits include:

- **Basis states** $|ijk\rangle$: 0-dimensional (fixed points under the maximal torus).
- **Product states**: 12-dimensional under local gates (the Segre variety of rank-1 tensors).
- **2-way entangled states**: 20-dimensional under one CNOT (the set of states reachable from a product state with A-B correlations only).
- **Fully entangled states**: 52-dimensional under $\mathfrak{su}(27)$ (all states in $\mathbb{C}P^{26}$ except a measure-zero subset).

5 Discrete Symmetry Group: $\mathrm{GL}(3,3)$

Beyond the continuous Lie algebra structure, the system possesses a discrete symmetry group that acts on the 27 computational basis states by permutation.

5.1 Generators

Consider the following discrete gates:

- $\mathrm{CNOT}_{i,j}$ (both directions) for each ordered pair of distinct qutrits $i, j \in \{A, B, C\}$.
- T_{inv} on each qutrit, acting as the inversion $|1\rangle \leftrightarrow |2\rangle$ with $|0\rangle$ fixed, i.e., the permutation (1 2) in the basis.

Proposition 9. *The group generated by CNOT gates (both directions) on all qutrit pairs and T_{inv} on each qutrit is isomorphic to* $\mathrm{GL}(3,3)$*, the general linear group over the field* $\mathrm{GF}(3)$*. Its order is*

$$|\,\mathrm{GL}(3,3)| = (3^3-1)(3^3-3)(3^3-3^2) = 11{,}232.$$

Proof. Identify the 27 basis states with the vector space $\mathrm{GF}(3)^3$ via the bijection

$$|ijk\rangle \leftrightarrow (i,j,k) \in \mathrm{GF}(3)^3,$$

where $\{0,1,2\}$ is identified with the elements of $\mathrm{GF}(3)$ (so that the group operation in $\mathrm{GF}(3)$ corresponds to addition modulo 3). The CNOT gate $\mathrm{CNOT}_{A,B}$ acts on the coordinates as

$$(i,j,k) \mapsto (i,\, i+j,\, k),$$

which is a linear transformation over $\mathrm{GF}(3)$ corresponding to the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathrm{GL}(3,3)$. Similarly, $\mathrm{CNOT}_{B,A}$ acts as $(i,j,k) \mapsto (i+j,\, j,\, k)$, and analogous formulas hold for the other wire pairs.

The T_{inv} gate on qutrit A acts as $(i,j,k) \mapsto (-i,j,k)$, i.e., the linear map $x \mapsto -x$ on the first coordinate (equivalently, multiplication by $2 \in \mathrm{GF}(3)^\times$). This is the diagonal matrix $\mathrm{diag}(-1,1,1) \in \mathrm{GL}(3,3)$ with determinant $2 \neq 1$, so it lies in $\mathrm{GL}(3,3) \setminus \mathrm{SL}(3,3)$. By the same token, T_{inv} on each qutrit gives the diagonal torus.

The six CNOT directions generate all elementary matrices $E_{ij}(1)$ (adding row j to row i), which in turn generate $\mathrm{SL}(3,3)$ over $\mathrm{GF}(3)$. Since $\mathrm{GL}(3,3) \cong \mathrm{SL}(3,3) \rtimes \mathrm{GF}(3)^\times$ with the determinant providing the splitting, and T_{inv} provides an element of determinant 2 (a generator of $\mathrm{GF}(3)^\times \cong C_2$), the combined group is the full $\mathrm{GL}(3,3)$.

The group acts transitively on the 26 non-zero vectors of $\mathrm{GF}(3)^3$ (the zero vector $|000\rangle$ is a fixed point). The stabilizer of any non-zero basis state is isomorphic to the affine group $\mathrm{GL}(2,3) \ltimes \mathrm{GF}(3)^2 \cong \mathrm{AGL}(2,3)$ of order $48 \cdot 9 = 432$, while the stabilizer of $|000\rangle$ is the full $\mathrm{GL}(3,3)$. □

Remark 10. *The Clifford group for three qutrits is strictly larger than* $\mathrm{GL}(3,3)$*: it includes the Hadamard gate and the phase gate, which introduce complex phases and do not act purely by permutations of basis states. The group* $\mathrm{GL}(3,3)$ *captures only the* permutation *symmetries of the computational basis under the restricted gate set of CNOT and inversion. Adding the generalized Pauli X gate (the cyclic shift $|i\rangle \mapsto |i+1 \bmod 3\rangle$) on each qutrit extends the group to the full affine group* $\mathrm{AGL}(3,3) = \mathrm{GF}(3)^3 \rtimes \mathrm{GL}(3,3)$*, as shown by Bocharov et al. [9][Appendix A].*

5.2 Relation to the Lie Algebra

The discrete group $\mathrm{GL}(3,3)$ is a subgroup of the permutation group S_{27} acting on the basis states (excluding the fixed point $|000\rangle$). Its linear action on $\mathbb{C}^{27}$ preserves the decomposition of $\mathbb{C}^{27}$ into the all-ones vector (the fixed point under $\mathrm{GL}(3,3)$) and its 26-dimensional orthogonal complement. The $\mathrm{GL}(3,3)$ action on this 26-dimensional subspace is not irreducible: it decomposes as a direct sum of three irreducible representations (of dimensions 1, 12, and 13).

The Lie algebra $\mathfrak{su}(27)$ contains the Lie algebra generated by the permutation matrices of $\mathrm{GL}(3,3)$. However, the continuous $\mathfrak{su}(27)$ is vastly larger: the discrete group

captures only the combinatorial symmetries of the basis, while the full Lie algebra generates continuous unitary rotations that map basis states to arbitrary superpositions.

6 Coupling Gate Classification

Proposition 11. *For $k = 2$, almost every $G \in \mathrm{SU}(9)$ has the property that the Lie algebra generated by $\mathfrak{su}(3)_A$, $\mathfrak{su}(3)_B$, and G (in the sense of Section 3.3) is $\mathfrak{su}(9)$. The exceptions are precisely those G belonging to the normalizer $N_{\mathrm{SU}(9)}(\mathrm{SU}(3) \times \mathrm{SU}(3))$, a proper closed algebraic subgroup (hence measure-zero in $\mathrm{SU}(9)$).*

Proof. Let $L_G = \mathrm{Lie}(\langle \exp(\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B), G \rangle)$. By Theorem 3, $L_G = \mathfrak{su}(9)$ iff L_G strictly contains $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$.

If G normalizes $\mathrm{SU}(3) \times \mathrm{SU}(3)$, then Ad_G preserves the subalgebra $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$, so L_G remains within this subalgebra. Conversely, if G does not normalize the subgroup, then $\mathrm{Ad}_G(X) \notin \mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$ for some X, and L_G strictly contains this 16-dimensional subalgebra. By Corollary 2, $\mathfrak{su}(3)_A \oplus \mathfrak{su}(3)_B$ is maximal in $\mathfrak{su}(9)$, so $L_G = \mathfrak{su}(9)$.

The normalizer is a proper closed algebraic subgroup of $\mathrm{SU}(9)$, hence a null set with respect to Haar measure. Its identity component is $\mathrm{SU}(3) \times \mathrm{SU}(3)$ itself; discrete components include the Weyl group element SWAP (which exchanges the two tensor factors).

Concrete exceptions include:

1. Tensor products $U_A \otimes U_B$ for any $U_A, U_B \in \mathrm{SU}(3)$.
2. The SWAP gate, which exchanges the two $\mathbb{C}^3$ factors and lies in the discrete part of the normalizer.
3. Diagonal matrices contained in the maximal torus of $\mathrm{SU}(3) \times \mathrm{SU}(3)$.

The CNOT gate is not a tensor product (as shown in the proof of Theorem 3), confirming that CNOT is a maximally entangling gate in the sense of generating the full algebra with a single application. □

7 Orthonormal Basis of $\mathbb{C}^{27}$

7.1 Basis Properties

The 27 basis states $|ijk\rangle$ form the standard orthonormal basis of $\mathbb{C}^{27}$. As a set of vectors in $\mathbb{C}^{27}$:

Vectors:	27
Pairwise inner products:	0 (all orthogonal)
Norm:	1 (unit vectors)

7.2 Frame Properties

The 27 basis vectors form a Parseval frame: the frame operator $F = \sum_i |e_i\rangle\langle e_i| = I_{27}$, i.e., the Bessel constant is 1 and the reconstruction formula is

$$|\psi\rangle = \sum_{i=1}^{27} \langle e_i, \psi \rangle |e_i\rangle$$

for all $|\psi\rangle \in \mathbb{C}^{27}$. Equivalently, the set is an orthonormal basis (the simplest nontrivial Parseval frame).

Under realification $\mathbb{C}^{27} \cong \mathbb{R}^{54}$, the 27 basis vectors map to 54 real vectors (each complex basis vector contributes two real dimensions). The resulting real spherical code on S^{53} has 54 points with pairwise inner products ± 1 for the real and imaginary parts coupled appropriately.

7.3 Capacity Comparison

The accessible information of 27 orthogonal states in $\mathbb{C}^{27}$ is $\log_2 27 \approx 4.755$ qubits per channel use. By comparison, three qubits (8 orthogonal states) have capacity $\log_2 8 = 3$ qubits per channel use. The ratio

$$\frac{\log_2 27}{\log_2 8} = \frac{4.755}{3.000} \approx 1.585$$

is precisely the per-symbol information advantage of qutrits over qubits, reflecting the $\log_2 3$ information content per qutrit.

7.4 Equiangular Lines

The 27 basis states form 27 mutually orthogonal lines through the origin in $\mathbb{C}^{27}$. The Gerzon bound for equiangular lines in $\mathbb{C}^{27}$ is

$$M \leq 27^2 = 729,$$

while our construction gives $M = 27$ (all pairwise inner products $= 0$, not constant). The maximal set of equiangular lines in $\mathbb{C}^{27}$ is a major open problem in quantum information theory (the existence of SIC-POVMs in dimension 27 is unknown).

8 Quantum Measurement

8.1 Projective Measurement

Given a pure state $|\Psi\rangle \in \mathbb{C}^{27}$ with coefficients c_{ijk} ($i, j, k \in \{0, 1, 2\}$), the natural readout is projective measurement in the computational basis $\{|ijk\rangle\}$. The probability of obtaining outcome (i, j, k) is

$$p_{ijk} = |c_{ijk}|^2.$$

8.2 Winner-Take-All Measurement

In a quantum setting, the full projective measurement already provides complete information: a single shot yields a trit string (i, j, k) drawn from the distribution $p_{ijk} = |c_{ijk}|^2$, and subsequent rounds can reconstruct the c_{ijk} via tomography. The winner-take-all (WTA) rule is a classical post-processing step that discards joint information and decides per-qutrit marginals. It may be useful when only the marginal values (not the full correlations) are needed, or as a complexity-reducing approximation.

For each qutrit $w \in \{A, B, C\}$, the marginal probability distribution is

$$p_w(s) = \sum_{\text{other qutrits}} |c_{ijk}|^2,$$

and the WTA readout declares the measured value to be $\mathrm{argmax}_s\, p_w(s)$. Ties are broken arbitrarily (e.g., by picking the first maximal value in the ordered set $\{0, 1, 2\}$).

8.3 Properties

1. **Deterministic on basis states:** For $|\Psi\rangle = |i_0 j_0 k_0\rangle$, the marginals are $p_A(i_0) = 1$, $p_B(j_0) = 1$, $p_C(k_0) = 1$, so the WTA output is (i_0, j_0, k_0) with probability 1.

2. **Product states:** For $|\Psi\rangle = |a\rangle \otimes |b\rangle \otimes |c\rangle$ (with unit-norm vectors), the marginals factorize as $p_A(s) = |a_s|^2$, $p_B(s) = |b_s|^2$, $p_C(s) = |c_s|^2$, so the WTA picks the dominant value per qutrit independently.

3. **Uniform superposition:** For $|\Psi\rangle = \frac{1}{\sqrt{27}} \sum_{i,j,k} |ijk\rangle$, all marginals are uniform: $p_w(s) = 1/3$ for all w and s. The WTA output is essentially a uniformly random trit per qutrit (broken ties arbitrarily if the marginals are exactly equal; with any infinitesimal perturbation, the output is uniformly distributed over all 27 symbols).

4. **Nonlinearity:** The WTA is a nonlinear post-processing of the measurement probabilities. It is not a linear operator on $\mathbb{C}^{27}$, but rather a piecewise-constant function whose decision boundaries are the hyperplanes where two marginals on the same qutrit are equal.

8.4 Quantum Detection and Noise Threshold

In a quantum optical implementation, the state $|\Psi\rangle$ is subjected to photon loss and detector noise. A simple noise model is the depolarizing channel on each qutrit:

$$\mathcal{E}_p(\rho) = (1 - p)\rho + \frac{p}{3} I,$$

where p is the depolarizing probability. After the channel, projective measurement in the computational basis yields outcome probabilities p_{ijk}. The WTA post-processing then selects the most likely trit per qutrit.

For a basis state $|i_0 j_0 k_0\rangle$ transmitted through independent depolarizing channels on each qutrit, the per-qutrit error probability is $2p/3$ (the probability that the depolarized state yields the wrong outcome on a given qutrit). The WTA symbol error probability is

$$P_e^{(\mathrm{WTA})}(p) = 1 - \left(1 - \frac{2p}{3}\right)^3 = \frac{2p}{3}\left(3 - 2p + \frac{4p^2}{9}\right).$$

For $p \ll 1$, the leading-order behavior is $P_e \approx 2p$ (a linear penalty from the three qutrits). Reliable computation requires $p \ll 5 \times 10^{-4}$ for $P_e < 10^{-3}$.

In the more general setting of a quantum channel with Gaussian noise (thermal noise in the optical modes or detector dark counts), the threshold analysis parallels the classical AWGN case but with complex amplitudes. The effective dimension doubles (from 27 real components to 27 complex components = 54 real dimensions), improving the noise threshold accordingly.

9 Remaining Open Problems

1. **Optimal spherical code on S^{53}.** Under realification, $\mathbb{C}^{27} \cong \mathbb{R}^{54}$, and the 27 basis states with their quadrature components yield 54 real antipodal points. The optimal configuration for 54 points on S^{53} is an open problem, as is the question of whether larger spherical codes exist within the $\mathfrak{su}(27)$ orbit of a basis state.

2. **SIC-POVMs in dimension 27.** The existence of a symmetric informationally complete positive operator-valued measure (SIC-POVM) in $\mathbb{C}^{27}$ is an open problem in quantum information theory. The 27 basis states are trivially IC but not SIC (the inner products are 0, not constant). The maximal size of equiangular line sets in $\mathbb{C}^{27}$ lies between 27 and 729, with the true maximum unknown.

3. **Error-correcting codes in $\mathbb{C}^{27}$.** The full reachable state space is $\mathbb{C}P^{26}$ (all pure states of three qutrits). Can we design quantum error-correcting codes within this space that protect against realistic noise models (depolarizing, amplitude damping, correlated dephasing) while respecting the gate set constraints? The 27-dimensional Hilbert space supports quantum codes that protect $\log_2 27 \approx 4.755$ logical qubits.

4. **Physical implementation.** The three-qutrit gate set (local $\mathfrak{su}(3)$ unitaries + CNOT) is directly realizable in several platforms:

   a. **Photonics:** Three spatial modes of light, each supporting three orbital angular momentum (OAM) states ($\ell = -1, 0, 1$) or three polarization-path hybrid states. Linear optical circuits implement SU(3) transformations on each spatial mode [5], and the CNOT gate can be realized via interferometric cross-coupling or nonlinear optics.

   b. **Superconducting circuits:** Three transmon qutrits with tunable couplers implement the CNOT gate and arbitrary single-qutrit unitaries via microwave pulses [6].

   c. **Trapped ions:** Three Sr^+ ions each encode a qutrit in the $5S_{1/2}$ manifold, with $\mathfrak{su}(3)$ rotations realized by Raman transitions and CNOT via the Mølmer–Sørensen gate [7].

   d. **Metaplectic anyons:** The $SU(2)_4$ anyon model naturally supports ternary computational basis states, with CNOT arising from braiding [9, 12].

5. **Computational universality.** We have established universality in SU(27): any unitary transformation on $\mathbb{C}^{27}$ can be approximated arbitrarily well by our gate set. The Solovay–Kitaev theorem guarantees efficient approximation for any dense subgroup of a compact Lie group, but explicit gate counts and error bounds for the ternary qutrit system merit further study.

6. **Generalization to higher dimensions.** The spanning tree theorem (Theorem 7) shows that $\mathfrak{su}(3^k)$ is generated for any k. What about more general tensor product structures $\mathfrak{su}(d_1) \otimes \cdots \otimes \mathfrak{su}(d_k)$? Our reliance on Dynkin's classification suggests that the maximal subalgebra argument extends to any dimensions $d_i \geq 3$, but the details depend on the branching rules.

10 Conclusion

We have presented a comprehensive Lie-algebraic analysis of a ternary qutrit system whose state space is $\mathbb{C}^{27} = \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$. Our main results are:

1. **Lie algebra generation** (Theorems 3, 4, 7): Local $\mathfrak{su}(3)$ generators on each qutrit together with CNOT gates on edges forming a spanning tree generate the full Lie algebra $\mathfrak{su}(3^k)$. For three qutrits, two overlapping CNOTs suffice to generate $\mathfrak{su}(27)$ (dimension 728). The proof relies on Dynkin's classification of maximal subalgebras (Type III tensor product embeddings for Type A).

2. **Maximal subalgebra argument**: The embedding $\mathfrak{su}(m) \oplus \mathfrak{su}(n) \to \mathfrak{su}(mn)$ is a Type III maximal subalgebra for $m, n \geq 3$. This provides a clean proof of universality without explicit computation. For $k = 2$, $\mathfrak{su}(3) \oplus \mathfrak{su}(3)$ is maximal in $\mathfrak{su}(9)$; for $k = 3$, $\mathfrak{su}(9) \oplus \mathfrak{su}(3)$ is maximal in $\mathfrak{su}(27)$.

3. **Coupling gate classification**: Almost every $G \in \mathrm{SU}(9)$ generates the full $\mathfrak{su}(9)$ together with the local algebras; the exceptions are precisely those G that normalize the subgroup $\mathrm{SU}(3) \times \mathrm{SU}(3)$. The CNOT gate is maximally entangling in this sense.

4. **Discrete symmetry** (Proposition 9): The discrete group generated by CNOT gates and local inversions is $\mathrm{GL}(3, 3)$, of order 11,232, acting on the 27 basis states with a fixed point at $|000\rangle$ and a single transitive orbit on the remaining 26 states.

5. **Orthonormal basis**: The 27 computational basis states form the standard orthonormal basis of $\mathbb{C}^{27}$. Projective measurement in this basis, with winner-take-all post-processing, defines the natural readout.

6. **Open problems**: The maximal number of equiangular lines in $\mathbb{C}^{27}$ remains unknown (between 27 and 729). SIC-POVMs in dimension 27, quantum error correction tailored to the gate set, and physical implementation in photonic, superconducting, or trapped-ion platforms are directions for future work.

These results provide the minimal gate set required for universal quantum computing on $\mathbb{C}^{27}$ using only local $\mathfrak{su}(3)$ operations and ternary CNOT couplings. The framework establishes that the $\mathfrak{su}(27)$ Lie algebra is generated with $k-1$ CNOT gates, matching the theoretical minimum, and that the accessible state space expands from product states ($\mathbb{C}P^2 \times \mathbb{C}P^2 \times \mathbb{C}P^2$) to the full projective space $\mathbb{C}P^{26}$ as the number of CNOT gates increases. This result establishes qutrits as a viable platform for universal quantum computing with minimal hardware overhead: only $k - 1$ CNOT gates and local $\mathfrak{su}(3)$ operations are required to unlock the full computational power of k qutrits.

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